

Research papers

A feature reuse based multi-model fusion method for state of health estimation of lithium-ion batteries[☆]

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ABSTRACT

Precise estimation of lithium-ion batteries's state of health (SOH) is crucial for their safe and stable operation. The existing SOH estimation methods mainly focus on a single model with multi-features. Hence, the potential of each feature cannot be fully exploited. To improve the accuracy of prediction model, a feature reuse (FR) based multi-model fusion method is proposed in this paper. Firstly, four features are extracted by analyzing the battery aging process from multiple perspectives such as sample entropy and time (micro/macro), particularly a new feature is proposed to reflect changes in battery energy. The four health factors are as first-level input and second-level feature reuse. Secondly, the preliminary SOH predictions are generated respectively by using k-nearest neighbor regression, random forest regression, linear regression, and extremely randomized tree regression models with different feature inputs. Finally, the initial SOH predictions were fused with the input features from the first stage using a Bayesian linear regression algorithm. Inspired by the advantages of feature reuse method and multi-model fusion, the feature reuse method is applied for fusing Bayesian regression multi-model. To verify the effectiveness of the proposed model, comparative experiments are carried on the Oxford and CALCE battery degradation dataset. Comparing with the single model estimation and methods without feature reuse, the proposed method has better accuracy and stronger robustness in SOH estimation.

1. Introduction

Lithium-ion batteries are widely used because of their good performance [1–3]. The battery health state (SOH) is defined as the percentage between the current available capacity and the initial capacity of the battery. An effective SOH estimation method is particularly important, which will directly affect the battery's performance of the work process [4]. Unlike voltage and current, SOH cannot be measured by sensors. It can only be estimated quantitatively by observing and identifying the characteristic parameters in the battery combined with an algorithm. In recent years, model-based and data-based methods as two dominant technologies have been used to predict battery SOH, and new advances have been made to increase the accuracy and stability of prediction. In the model-based method, the focus is on the physical and chemical structure of the battery, which is expressed in mathematical equations. These models mainly consist of electrical equivalent circuit model (EECM), electrochemical model (ECM), and empirical model (EM) [5]. In these experiments, a leave-one-out cross-validation method was used, which take the data of one battery in the dataset as the test set, and the data of the remaining battery

as the training set. Based on the established EECM such as the RC model [6] and FOM [7], adaptive filtering algorithms, for example, EKF [8] and AEKF [9] are used to identify the electrical parameters such as resistance and capacity estimated by battery SOH. In terms of ECM, it considers the electrochemical reactions that cause the fading of the battery life. EM is another vital model to estimate SOH by building a relationship between the degradation factors and the battery SOH. The EECM-based approach considers the electrical characteristics of the battery. Thus, degradation features can be easily performed and extracted.

However, it is difficult to build an accurate battery model due to the complex internal principles of the battery and uncertain working conditions. Researchers are increasingly considering data-driven methods. In the data-driven methods, the battery SOH is estimated by mining the feature vectors during aging. There are two key aspects that affect the effectiveness of the method. The first aspect is whether the health features input to the model adequately describe the battery degradation, and the second aspect is whether the model training fully exploits the feature potential. Incremental capacity curve analysis (ICA) studies

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have been demonstrated to be available for online SOH estimation from various perspectives such as area, position and gradient [10–12]. In Ref. [13] several feature extraction algorithms based on IC curves were proposed. A Gaussian process regression model based on partial incremental capacity curves is proposed in Ref. [14]. Recently, Lin et al. [15] extracted the voltage-time difference from the voltage curve of the constant current charging (CCCT) stage as a health feature, and experiments showed that CCCT is an ideal substitute for the incremental capacity peak area. Widodo et al. [16] proposed a smart battery health prediction method (SampEn) based on the sample entropy of discharge voltage features. Zou et al. [17] extracted a new health metric from the discharge voltage to characterize the capacity of Li-ion batteries. Richardson et al. [18] used the value of time at equal voltage as an input to the GPR to estimate SOH.

Although the literature [10–18] has improved in terms of feature extraction or model, the features cannot comprehensively describe the battery aging process due to the analysis of predicting the battery SOH from only one perspective, which leads to the limitation of the accuracy and robustness capability of the model. To solve this problem, Ref. [19–21] extracted several features from temperature variation, voltage and current curves, and incremental rate curves to build SOH prediction models, respectively. In Ref. [22] the authors analyzed the voltage, current and temperature during the charging and discharging of Li-ion batteries from various perspectives and selected the health features reflecting the battery degradation as the input of the model for SOH prediction.

The methods in Ref. [19–22] are essentially based on the fusion of multiple features. Although multi-features can comprehensively describe the battery aging process, it is difficult for the model to fully exploit the potential of the features due to the limitation of the model capability.

In order to fully exploit the potential of each feature and ensure the overall performance of SOH estimation, the method of multi-model fusion is applied to this field. Ma et al. [23] extracted features from different views during the discharge process of lithium-ion batteries, and developed a multiple-view feature fusion with a support vector regression (SVR) ensemble strategy (MVFF-ESVR). Tian et al. [24] proposed to fuse the ICA curve prediction results and the temperature curve prediction results using a Kalman filter framework, respectively. Liu et al. [25] used the idea of superposition algorithm to estimate battery SOH by fusing different characteristic parameters, combining support vector regression (SVR) and extended Long-Short Term Memory (LSTM). Lin et al. [26] developed a multi-model fusion model based on random tree regression. Unfortunately, they suffer from complex fusion models, difficult hyperparameter determination, and limited accuracy.

To address the above limitations, in this paper, a feature reuse-based multi-model fusion method is proposed for Li-ion battery state-of-health estimation. The aging features are extracted from the aspects of energy, sample entropy and time(micro/macro) to describe the aging behavior more comprehensively. Due to the excellent performance of machine learning methods, K-Nearest Neighbor Regression(KNNR) [27], Extremely Randomized Trees Regression(ERT) [28], Logistic Regression(LR) [29] and Random Forest Regression(RFR) [30] models are initially used to predict SOH. Inspired by the advantages of the feature reuse method and multi-model fusion, the feature reuse method is applied to Bayesian Linear Regression(BLR) [31] multi-model fusion. The results show that our proposed method exhibits better accuracy and stronger robustness in SOH prediction.

The paper is organized as follows. Section 2 gives how health features extracted from four perspectives: time (micro-macro), sample entropy, and energy. Section 3 illustrates the multi-model fusion approach based on feature reuse implementation. The experiments and analysis are performed in Section 4. Finally, conclusions are given in Section 5.

Table 1

The overview of two battery datasets.

Dataset	Oxford	CACLE CS2
Number of cells	8	4
Form factor	Pouch	Prismatic
Cathode	LiCoO2 /LiNiMnCoO2	LiCoO2
Capacity rating	740 mAh	1100 mAh
Voltage ranges	2.7 V–4.2V	2.7V–4.2V
Depth of discharge	27.5% ± 0.5% To 21% ± 1%	100% To 69% ± 6%
Charge	CC (2C)	CC-CV (0.5C)
Discharge	Artemis urban driving cycle	CC (1C)
Temperature	40°C	25°C

Note: CC stands for constant current charging and CV stands for constant voltage charging.

2. Data analysis and feature extraction

2.1. Data set

The Oxford and CALCE datasets were used to validate the model in this paper. For the Oxford dataset [18], batteries with serial numbers Cell 1 to Cell 8 were selected, and for the CALCE dataset [32], batteries with serial numbers CS-35 to CS-38 were selected; an overview of the two datasets is given in Table 1.

2.2. Energy feature extraction

Fig. 1 shows the discharge voltage curves for the constant current phase of the battery in different cycle numbers. For the different number of cycles, the battery takes different time to complete the constant current discharge process. The result is a difference in the amount of energy released in each cycle. Traditional energy-based methods characterize the battery aging process by directly calculating the constant current discharge phase energy according to Eq. (1).

$$Energy_i = \int_{t_0}^{t_{cut}} V_i \cdot I_i(t) dt \quad (1)$$

where t_0 and t_{cut} are start and cut-off time of the discharge, V_i and I_i represent the voltage and current at cycle i respectively.

However, in the actual test, the current may fluctuate much due to equipment and other reasons that will lead to errors. Therefore, a new health feature is proposed in this paper that can reflect the variation of battery energy with the number of cycles.

Partial integration(partial integration of voltage in time) is as a new feature, which is defined as the integration in time of the difference between the discharge voltage and the cut-off voltage. As shown in Fig. 1, this region decreases as the battery age.

The extraction process of partial integration of voltage in time (PIVT) includes three main steps as follows:

- Step1 Extraction of the voltage and time data for each cycle in a constant current discharge stage;
- Step2 The algebraic relationship between discharge voltage and time is fitted by using six-order polynomial as Eq. (2);

$$V_i = At_i^6 + Bt_i^5 + Ct_i^4 + Dt_i^3 + Et_i^2 + Ft_i + G \quad (2)$$

where, V_i and t_i are the voltage and time of the i th cycle respectively. Parameters A, B, C, D, E, F and G are determined by the 'polyfit' function in Matlab.

- Step3 Calculating the integral according to Eq. (3).

$$PIVT_i = \int_{t_0}^{t_{cut}} V_i - V_s dt \quad (3)$$

where V_i and V_s represent the start and cut-off voltages at cycle i respectively.

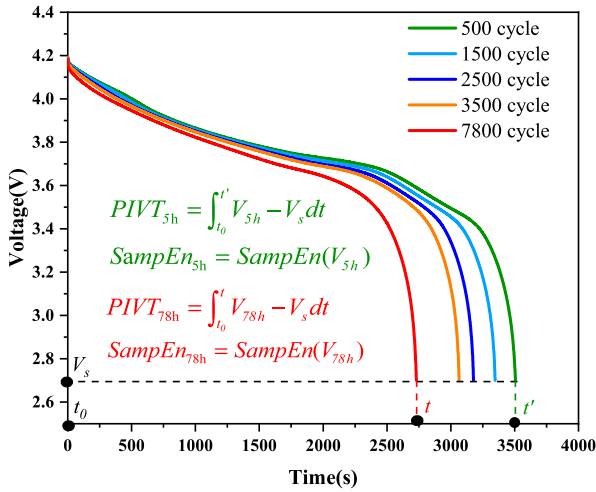


Fig. 1. Discharge voltage change curves in different cycles of the Cell 1.

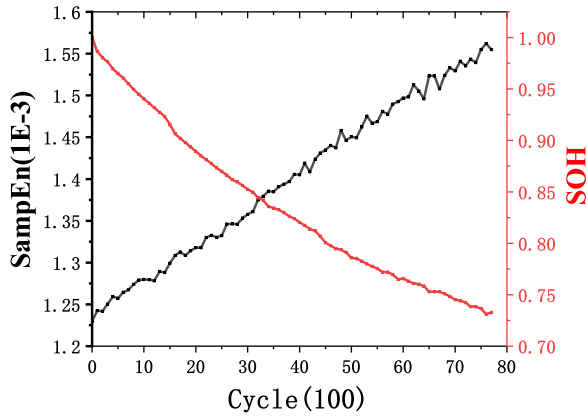


Fig. 2. Sample entropy and SOH in different cycles of the Cell 1.

2.3. Sample entropy feature extraction

Sample entropy is a valuable tool for signal complexity and predictability analysis. As the battery fades, the measurable voltage response under the same excitation changes accordingly. The salient feature of this method is that the SOH is estimated at the end of each discharge cycle simply by calculating the voltage's sample entropy, which does not involve a complex computational process and avoids the cumulative error caused by integration. Based on the above advantages, the sample entropy features are widely used to describe the battery aging process.

Calculate the sample entropy of the discharge voltage in the constant current phase as a healthy feature. In this paper, the sample entropy is calculated according to Eqs. (4)–(8), and then the sample entropy curve and the SOH curve are shown in Fig. 2. According to the figure, the sample entropy of the battery increases with the cycle number during the discharge process. Thus, this feature can describe the process of battery aging.

Sample entropy algorithm flow:

- Step1 Get the time series data under N sampling points, denoted as $x(1), x(2), \dots, x(N)$.
- Step2 Define the comparison vector m with the similarity measure r .
- Step3 Define the comparison vector:

$$X_m(i) = \{x(i), x(i+1), \dots, x(i+m-1)\},$$

$$1 \leq i \leq N - m + 1, i \in N \quad (4)$$

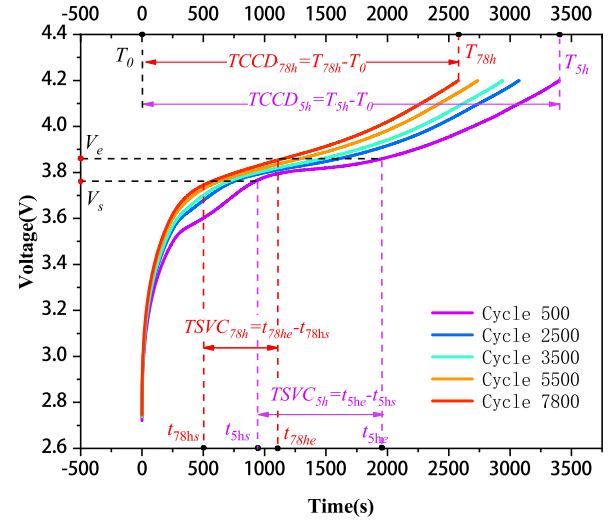


Fig. 3. Charging voltage curves in different cycles of the Cell 1.

Step4 Define the distance difference of the vectors:

$$d(X_m(i), X_m(j)) = \max_{k \in [0, m-1]} (|X_m(i+k) - X_m(j+k)|),$$

$$0 \leq i \leq N - m + 1, j \neq i, j \in N \quad (5)$$

Step5 Calculate the number of vector distances that satisfy $d(X_m(i), X_m(j)) \leq r$, denoted as B_i .

Step6 Calculate the proportion of B_i to the total number of comparison vectors, denoted as $B_i^m(r)$:

$$B_i^m(r) = \frac{B_i}{N - m + 1} \quad (6)$$

Step7 Calculate the average value of $B_i^m(r)$:

$$B_m(r) = \frac{1}{N - m} \sum_{i=1}^{N-m} B_i^m(r), \quad 0 \leq i \leq N - m + 1 \quad (7)$$

Step8 Increase the length of the comparison vector to $m+1$ and repeat steps 3–8 to calculate $B_{m+1}(r)$.

Step9 Calculate the sample entropy:

$$\text{SampEn}(m, r, N) = -\ln \left[\frac{B_{m+1}(r)}{B_m(r)} \right] \quad (8)$$

2.4. Time feature extraction

Time features have been widely studied for their advantages, such as easy extraction and strong interpretation. Currently, two types of time features are classified in the constant current charging stage; macroscopic time features for the entire charging stage in each cycle and microscopic time features for the time difference corresponding to a specific voltage segment of the charging stage in each cycle. In order to more accurately describe the battery aging process, time features are extracted from different scales during the constant current charging stage in this paper.

Choose time of constant current mode duration (TCCD) as a macro time feature. According to Fig. 3, the constant-current charging time decreases as the number of cycles increases. The definition of the TCCD is the time elapsed from a certain starting voltage for constant current charging until a certain cut-off voltage. (Starting voltage: 2.7 V, Cut-off voltage: 4.2V). Most of the power will be charged to the battery at this stage. The duration directly indicates how much capacity can be charged in the constant current mode, which reflects the battery polarization phenomenon. As the number of cycles increases, the battery

Table 2

Pearson correlation coefficient of the extracted health factors.

Cell	Pearson correlation coefficient			
	TCCD	SampEn	TSVC	PIVT
Ce11 1	0.9999	-0.9853	0.9978	0.9998
Ce11 2	0.9999	-0.9839	0.9973	0.9999
Ce11 3	0.9999	-0.9876	0.9982	0.9989
Ce11 4	0.9999	-0.9885	0.9974	0.9999
Ce11 5	0.9104	-0.9133	0.9863	0.9999
Ce11 6	0.9999	-0.9848	0.9976	0.9991
Ce11 7	0.9999	-0.9896	0.9985	0.9999
Ce11 8	0.9999	-0.9908	0.9987	0.9999

gradually ages, and the polarization phenomenon gradually increases, so the TCCD gradually decreases.

Choose time difference corresponding to the specific voltage band of the charging phase (TSVC) as a micro time feature. The voltage segment selection range was obtained from the [15]. Specifically, the voltage ranges [3.77 V, 3.87V] and [3.87 V, 3.97V] were selected for the Oxford and CALCE datasets, respectively. The literature found that the voltage in this fixed voltage range corresponds to the area of the main peak of the IC curve. It is demonstrated that this feature can replace the IC peak feature.

Based on the above analysis, it can be seen that the variation of both TCCD and TSVC with cycling can characterize the battery SOH. The TCCD and TSVC feature extraction methods are shown in Fig. 3.

2.5. Correlation analysis

In order to prove the effectiveness of extracting health factors, Pearson correlation coefficient is used to analyze the extracted health factors. As shown in Table 2. Pearson correlation coefficient formula is as Eq. (9). Suppose there are two arrays x and y , the correlation coefficient r_{xy} , and the corresponding relationship is shown below. The closer the absolute value of correlation coefficient is to 1, the higher the correlation is. It can be seen from the table that the extracted health features have a high correlation with the real SOH value, reaching more than 0.9, with a strong correlation. Therefore, the extracted health features can be used as the input of the prediction model.

$$r_{xy} = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{\sqrt{n \sum x_i^2 - \sum x_i^2} \sqrt{n \sum y_i^2 - \sum y_i^2}} \quad (9)$$

where n is the number of observation objects, x_i is the observation value of each x , and y_i is the observation value of each y .

3. Methodologies of the multi-model fusion

To overcome the problems of under-exploitation of feature potential and poor prediction performance, feature reuse is used in multi-model fusion features for more accurate and robust results. First, battery energy, sample entropy and time(micro/macro) are extracted from the test data and inputted respectively to four different models. KNNR, ERTR, LR and RFR are used to predict SOH. The single-model prediction results and the four features are all inputted to the multi-model fusion. Finally, the Bayesian Linear Regression(BLR) [31] method is used to fuse multiple models. The multi-model fusion algorithm flow based on feature reuse is shown in Fig. 4.

3.1. KNNR prediction

The idea of KNN regression is to find the nearest K known samples to the sample for testing and make their mean value as sample value to be tested. For example, assume that a_1, a_2, a_k are the k nearest neighbor features of respective x , and their target values are $V_1, V_2 \dots V_k$, then the values of the target x are as follows:

$$V_x = \frac{1}{k} \sum V_i \quad (10)$$

where k is the window size.

3.2. ERTR prediction

ERTR takes all samples as the training set, CART decision tree as the base machine model, and finally outputs the results based on the weak learner results. The Eq. (11) is the result obtained by integrating all the ET models and using the mean value method.

$$\hat{y} = \sum_{x=1}^T (f_x / T) \quad (11)$$

where y is the prediction result, f_x is the prediction result of the x decision tree.

3.3. LR prediction

Assume that SOH is linearly related to the input feature x as follows.

$$h_1(x) = x\beta + \varepsilon \quad (12)$$

where x is the regression coefficient, ε is the random error with zero mean.

3.4. RFR prediction

Assume Y is the input of RFR. The principle of RFR is simplified: firstly, the Out-of-bag (OOB) data is generated using Bootstrap. Secondly, each node completes the data collection and splitting. Finally, a mode is built to predict the OOB error, and the output value is obtained depending on the result. The expression of the RFR prediction model is shown in Eq. (13).

$$H = \arg \max \sum_{i=1}^N (K(h(Y, \Theta_i)j)) \quad (13)$$

where H is the prediction result, K is the schematic function, h is the prediction result of the decision tree, j is the constant.

3.5. BLR fusion algorithm

BLR algorithm uses probability distributions rather than point estimates to construct linear regressions. Since BLR is highly adaptive to the data and has no hyper-parameters, it can provide high predictive efficiency. A BLR model was selected as the fusion model based on the comparison results of this feature and other models. The feature reuse based multi-model fusion method is divided into two specific steps as follows.

- Step1 Define R_1 as the prediction of the TCCD input to KNNR.
Define R_2 as the prediction of the SampEn input to ERTR.
Define R_3 as the prediction of the TSVC input to LR.
Define R_4 as the prediction of the PIVT input to RFR.
- Step2 The prediction results and health features are composed of an 8-dimensional vector $x = \{R_1, R_2, R_3, R_4, \text{TCCD}, \text{SampEn}, \text{TSVC}, \text{PIVT}\}$ as input to the BLR.

The specific details of the BLR algorithm are as follows. Multiple linear regression elaborates the model of variability between the corresponding variable (y) and the explanatory variable x . expressed as:

$$y_t = \beta x_t + \varepsilon_t, t = 1, 2, 3, \dots \quad (14)$$

where β is the regression coefficient, ε_t is the regression disturbance having a mean of 0 and a variance of σ^2 .

In the BLR approach ε_t and σ^2 are random variables. In short, the probability distribution of the parameters is updated with the likelihood function and the prior distribution. The multiple linear regression likelihood for a random sample is expressed as:

$$(\beta, \sigma^2 | y, x) = \prod_{t=1}^T P(y_t | x_t, \beta, \sigma^2) \quad (15)$$

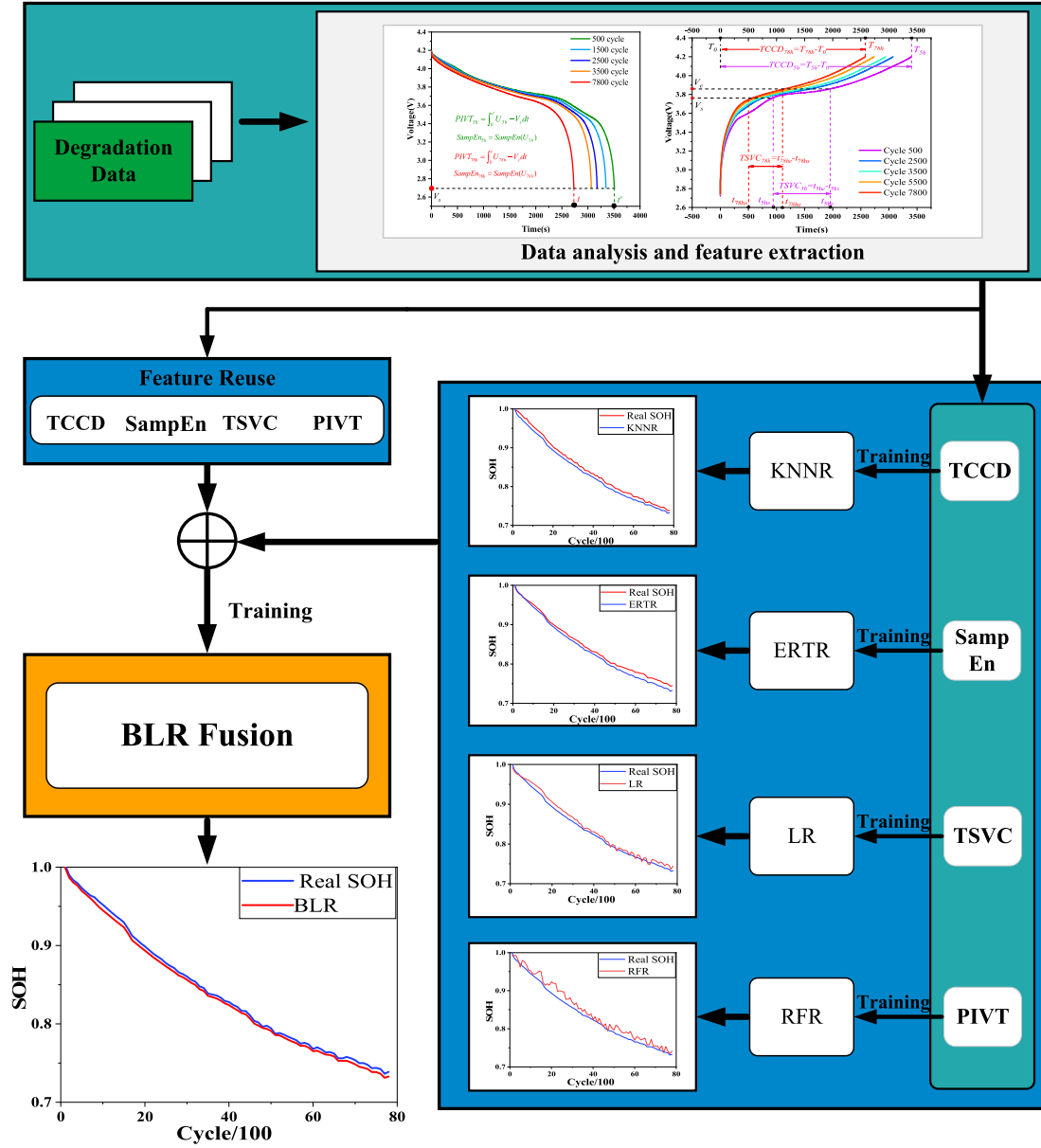


Fig. 4. The flow of feature reuse based multi-model fusion algorithm.

where $P(y_i|x_i, \beta, \sigma^2)$ is the conditional probability density function of y_i .

If the disturbances are independent, homoscedastic and Gaussian, then

$$(\beta, \sigma^2 | y, x) = \prod_{i=1}^T \phi(y_i; x_i \beta, \sigma^2) \quad (16)$$

where $\phi(y_i, x_i, \beta, \sigma^2)$ is the Gaussian probability density evaluated at y_i with mean $x_i \beta$ and variance σ^2 , $\pi(\beta)$ and $\pi(\sigma^2)$ are parametric prior distributions, respectively. Using Bayes' rule, it can be obtained:

$$\begin{aligned} \pi(\beta, \sigma^2 | y, x) &= \frac{\pi(\beta) \pi(\sigma^2) \ell(\beta, \sigma^2 | y, x)}{\int_{\beta, \sigma^2} \pi(\beta) \pi(\sigma^2) \ell(\beta, \sigma^2 | y, x) d\beta d\sigma^2} \\ &\propto \pi(\beta) \pi(\sigma^2) \ell(\beta, \sigma^2 | y, x) \end{aligned} \quad (17)$$

The denominator which is a constant is the distribution of the response at a given predicted value. From the data, the expectation of $\hat{\beta}$ is:

$$\hat{\beta} = E(\beta | y, x) = \int_{\beta, \sigma^2} \beta \pi(\beta, \sigma^2 | y, x) d\beta d\sigma^2 \quad (18)$$

Given the data, $\hat{y}$ is predicted by $\hat{x}$ based on (19).

$$\pi(\hat{y} | y, x, \hat{x}) = \int_{\beta, \sigma^2} f(\hat{y} | \beta, \sigma^2 \hat{x}) \pi(\beta, \sigma^2 | y, x) d\beta d\sigma^2 \quad (19)$$

The outcome can be considered as the conditional expected value of the probability distribution of y concerning the posterior distribution of the parameters.

4. Experiments and analysis

In this paper, SOH is the ratio of the current maximum capacity to the initial capacity as shown in Eq. (20).

$$SOH = Q_{\text{current}} / Q_{\text{initial}} \quad (20)$$

where Q_{current} is the current maximum capacity during the cycle and Q_{initial} is the initial capacity.

The model's effectiveness is validated on Oxford and CALCE battery degradation datasets. The experiments are divided into two parts: comparison single model with single feature and comparison model

Table 3
RMSE and MAE of SOH estimation of different models on Oxford and CALCE datasets.

Battery	RMSE					MAE				
	RFR	ERTR	LR	KNNR	BLR-FR	RFR	ERTR	LR	KNNR	BLR-FR
Cell 1	1.43%	0.91%	0.84%	0.95%	0.44%	1.43%	0.84%	0.77%	0.93%	0.43%
Cell 2	1.32%	1.18%	1.20%	0.94%	0.92%	0.80%	0.53%	0.94%	0.52%	0.77%
Cell 3	1.31%	0.39%	0.49%	0.28%	0.25%	0.79%	0.31%	0.40%	0.24%	0.20%
Cell 4	0.80%	0.43%	0.82%	0.34%	0.24%	0.65%	0.39%	0.71%	0.29%	0.22%
Cell 5	0.91%	0.47%	0.46%	0.46%	0.25%	0.75%	0.38%	0.39%	0.44%	0.20%
Cell 6	0.82%	0.32%	0.42%	0.58%	0.16%	0.67%	0.28%	0.35%	0.43%	0.14%
Cell 7	0.90%	0.89%	1.24%	0.69%	0.18%	0.73%	0.66%	1.09%	0.67%	0.16%
Cell 8	1.28%	1.25%	0.91%	0.95%	0.43%	1.04%	0.93%	0.81%	0.93%	0.38%
CS-35	1.22%	1.86%	1.50%	1.30%	0.80%	0.93%	1.54%	1.18%	0.99%	0.60%
CS-36	1.24%	1.64%	1.70%	1.44%	0.98%	0.98%	1.34%	1.28%	1.19%	0.82%
CS-37	1.03%	1.77%	1.02%	1.21%	0.73%	0.78%	1.44%	0.80%	0.94%	0.58%
CS-38	1.22%	1.69%	1.89%	2.00%	0.84%	0.91%	1.30%	1.20%	1.39%	0.63%

fusion of without feature reuse. In these experiments, a leave-one-out cross-validation method was used, which take the data of one battery in the dataset as the test set, and the data of the remaining battery as the training set. To evaluate the prediction model's performance, mean absolute error (MAE) and root mean square error (RMSE) were used to predict SOH in this study. The standard descriptions are as follows:

$$MAE = \frac{1}{m} \sum_{i=1}^m |\hat{Y}(i) - Y(i)| \quad (21)$$

$$RMSE = \sqrt{\frac{1}{m} \sum_{i=1}^m (\hat{Y}(i) - Y(i))^2} \quad (22)$$

where $Y(i)$ and $\hat{Y}(i)$ represent the measurement capacity series and predicted capacity, and i implies the number of cycles.

4.1. Comparison single model with single feature

4.1.1. Oxford dataset

To verify the validity of the proposed model, the output results of the first stage using a single feature and a single model are compared with the results of the second stage using multi-model fusion based on feature reuse. In Table 3, BLR-FR represents the results of the multi-model fusion method using feature reuse, and the rest are the results of the first stage output. The estimation results are shown in Fig. 5. These results show that the proposed fusion model can obtain smooth tracking performances, which can intuitively show the most realistic reflection of the battery degradation trend. As shown in Fig. 5, the maximum absolute error (MAX) of fusion model on Cell 1 and Cell 3 to 8 are less than 1%. It should be noted that the maximum error of RFR and ERTR prediction on Cell 2 is close to 8%, and the maximum absolute error after fusion is only about 2%, which reduces the error by 75%. The proposed fusion method can effectively reduce the maximum error. The KNNR can predict the SOH of the battery well, but the fluctuation of the SOH prediction error gradually increases with the aging of the battery. The estimation performance of RFR is insufficient because there is a large bias in the first half of the cycle. LR performance gets progressively worse with cyclic estimation. ERTR is easy to have large errors where SOH varies a lot. It can be seen by comparison that, the proposed fusion model with the introduction to feature reuse can describe the battery aging process well and provide higher accuracy and robustness for the SOH estimation.

The MAE, RMSE of the estimation results of all eight cells are summarized in Table 3. It can be seen from the table, the RMSE of all Cells reached the minimum level compared to the single model, especially Cell 5 to 8, with RMSE of 0.25%, 0.16%, 0.18% and 0.43% respectively, which were 45.65%, 50%, 73.91% and 52.75% lower than the minimum error of the single model, respectively.

Analysis in terms of MAE, the MAE of the proposed fusion model on Cell 1 and Cell 3 to 8 are all less than 0.5%. Especially, The MAE of cell

5 to 8 were 0.20%, 0.14%, 0.16%, and 0.38%, respectively, which were 48.72%, 50%, 75.76%, and 53.09% lower than the minimum error of a single model, respectively. The results show that multi-model fusion can effectively reduce the MAE, and SOH can be accurately predicted in most cases.

4.1.2. CALCE dataset

To further verify the validity of the proposed method, CALCE dataset validation is used to test. The prediction results using the CALCE dataset are shown in Table 3. By the fusion model, the RMSE and MAE of all Cells reached the minimum level compared to the single model. The RMSE and MAE of CS-35 to 38 are within 1%. Specifically, The RMSE of CS-35 and 38 are 0.80% and 0.84% respectively, which are 34.43% and 31.15% lower than the lowest errors in the other four models. The MAE for CS-36 and 37 are 0.82% and 0.58% respectively, which are 16.33% and 26.64% lower than the lowest errors in the other four models, respectively. The result shows that the proposed method is very adaptable and can effectively predict SOH.

4.2. Comparison model fusion of without feature reuse

To verify the effectiveness of the feature reuse method and the advantages of the BLR-FR model, the results of multi-model fusion using RFR and BLR in two cases (with and without feature reuse) are compared using the first level output from the previous section as input. The abbreviations are RFR-WFR, RFR-FR, BLR-WFR and BLR-FR. Fig. 6 and Table 4 show the results of the above comparison experiments. It can be seen that the RMSE and MAE decrease when the FR method is added to the model. The prediction results have good agreement and smoothness with the real SOH. The average RMSE of the RFR fusion model and BLR fusion model using FR are 0.63% and 0.35%, which was 21.25% and 52.7% respectively lower than without FR. The RMSE of Cell 6 and 7 with RFR-FR method are 0.35%, and 0.69% respectively, which are 36.36%, 27.37% lower than the without FR method. Especially, in the BLR fusion, the RMSE for Cell 1 and Cell 6 to 8 with FR method are 0.44%, 0.16%, 0.18% and 0.43% respectively, which are 58.1%, 67.35%, 81.82% and 64.46% lower than the without FR method over the BLR fusion. The average MAE of the RFR fusion model and BLR fusion model using FR are 0.51% and 0.31%, which are 22.73% and 46.88% lower than without FR. Among them, the average errors of Cell 7 and 8 are 0.65% and 0.87% after using the FR method, compared with 25.28% and 20.18% without the FR method. It is noteworthy that the MAE for Cell 6 to 8 with BLR-FR are 0.14%, 0.16%, 0.38% respectively, which are 67.35%, 81.82% and 64.46% lower than the without FR method over the BLR fusion. These results show that the FR method can effectively reduce the single model prediction error, and improve the estimation accuracy of model fusion. Then, compared the prediction results of RFR-FR with BLR-FR, it can be observed that the accuracy of BLR-FR estimation results is more accuracy (Excluding Cell 2). Specifically, the RMSE of Cell 1 and Cell 6 to 8 with BLR-FR are

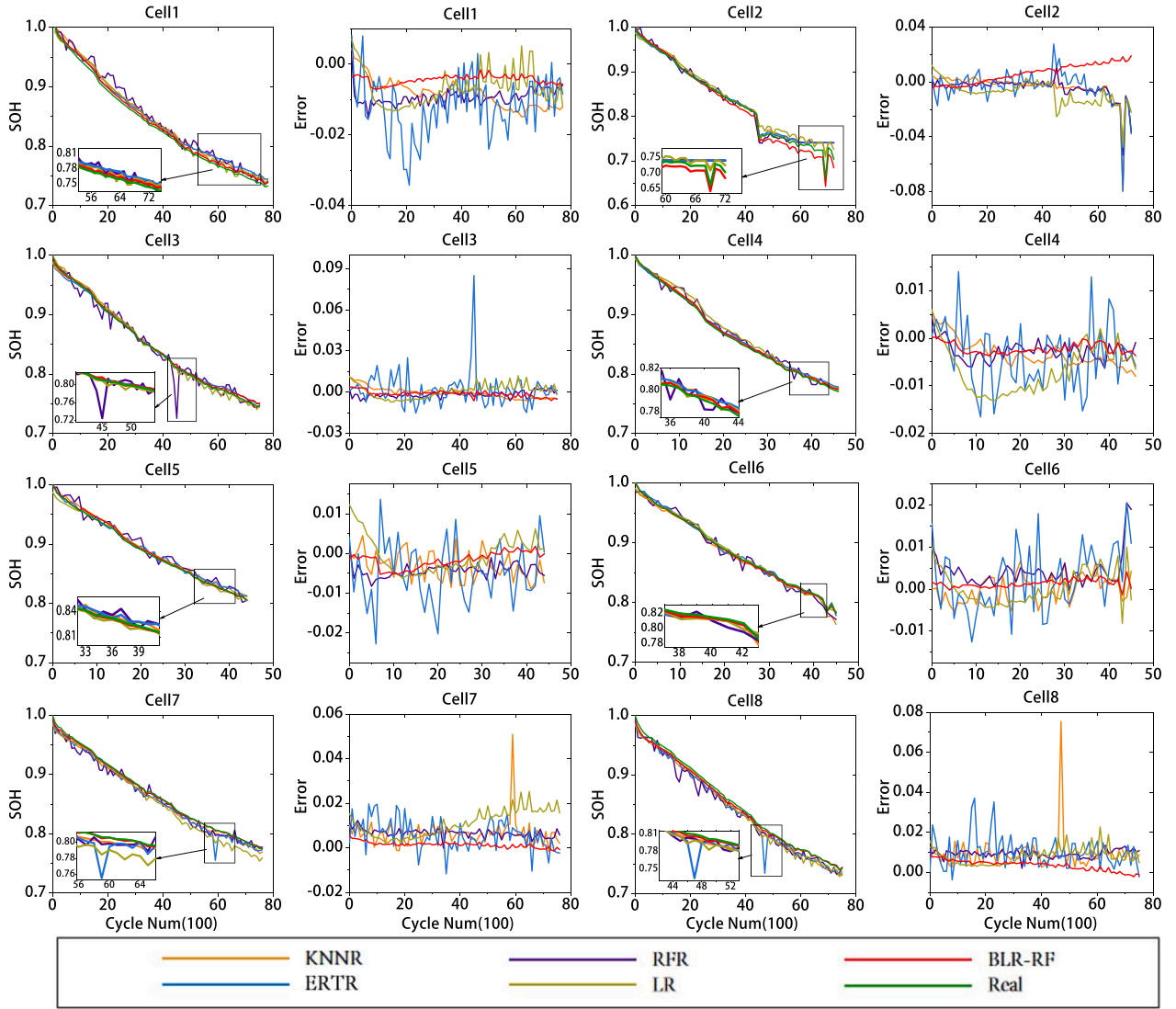


Fig. 5. SOH estimation results of different models on Oxford battery dataset.

Table 4
RMSE and MAE of SOH estimation results of different fusion methods on Oxford battery dataset.

Cell	RMSE				MAE			
	RFR-WFR	RFR-FR	BLR-WFR	BLR-FR	RFR-WFR	RFR-FR	BLR-WFR	BLR-FR
1	1.08%	0.84%	1.05%	0.44%	1.01%	0.81%	1.03%	0.43%
2	1.19%	1.07%	1.05%	0.92%	0.65%	0.45%	0.65%	0.77%
3	0.37%	0.28%	0.28%	0.25%	0.32%	0.22%	0.23%	0.20%
4	0.60%	0.55%	0.44%	0.24%	0.54%	0.50%	0.40%	0.22%
5	0.51%	0.41%	0.46%	0.25%	0.42%	0.35%	0.43%	0.20%
6	0.55%	0.35%	0.49%	0.16%	0.42%	0.27%	0.36%	0.14%
7	0.95%	0.69%	0.99%	0.18%	0.87%	0.65%	0.95%	0.16%
8	1.18%	0.91%	1.21%	0.43%	1.09%	0.87%	1.14%	0.38%

0.44%, 0.16%, 0.18% and 0.43% respectively, which are 0.40%, 0.19%, 0.51% and 0.48% lower than the RFR-FR results. The MAE of Cell 1 and Cell 4 to 8 with BLR-FR are 0.43%, 0.22%, 0.20%, 0.14%, 0.16% and 0.38% respectively, which are 0.38%, 0.28%, 0.15%, 0.13%, 0.49% and 0.49% lower than the RFR-FR results. This shows that BLR-FR has a higher prediction performance than RFR-FR fusion. In conclusion, the FR method can effectively reduce the error, and the BLR-FR method can provide satisfactory and robust prediction performance for battery health conditions.

4.3. Comparison with other model

The algorithm studied in this paper is essentially a stacking algorithm, so it is necessary to compare it with similar advanced algorithms. The algorithm proposed in this paper is compared with the multi-feature-based multi-model fusion method mentioned in the literature. The comparison results of the above prediction methods are shown in Table 5.

Based on the comparison of the prediction results in Table 5, it can be seen that the root mean square error of the BLR-FR algorithm for

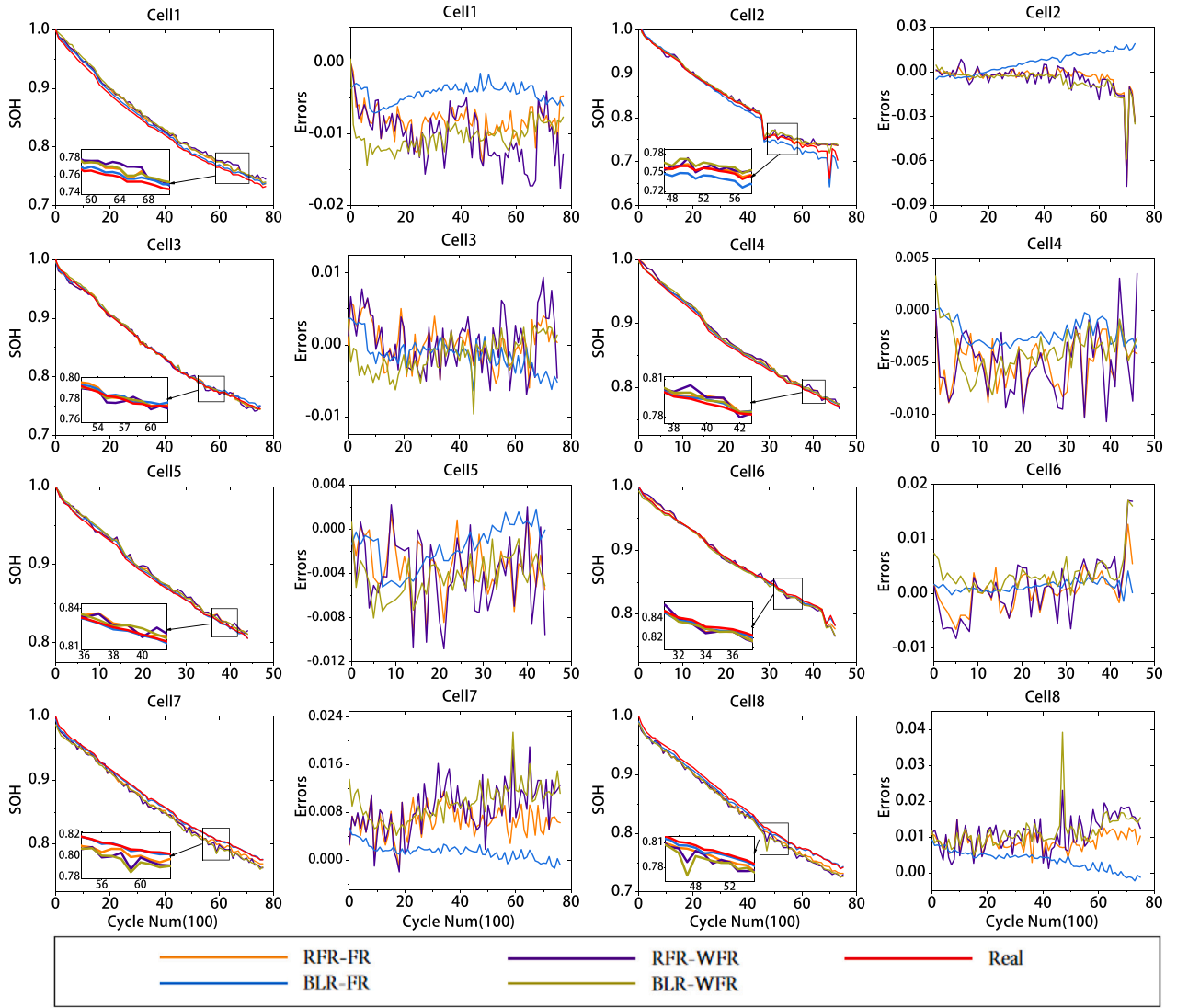


Fig. 6. SOH estimation results of different fusion methods on Oxford battery dataset.

Table 5
Comparison with other model.

Cell	RMSE	
	BLR-FR	Other model [26]
Cell 1	0.44%	1.10%
Cell 2	0.92%	1.26%
Cell 3	0.25%	0.39%
Cell 4	0.24%	0.51%
Cell 5	0.25%	0.47%
Cell 6	0.16%	0.35%
Cell 7	0.18%	0.64%
Cell 8	0.43%	1.06%

cells 1-8 is 0.44%, 0.92%, 0.25%, 0.24%, 0.25%, 0.16%, 0.18%, 0.43%, and the results of the multi-feature-based multi-model fusion model are 1.10%, 1.26%, 0.39%, 0.51%, 0.47%, 0.35%, 0.64%, and 1.06%. The proposed algorithm outperforms the multi-feature-based multi-model fusion algorithm in predicting the results on 8 cells. Therefore, the BLR-FR algorithm has higher accuracy.

The multi-model fusion method based on feature reuse can fully exploit the feature potential and effectively improve the prediction accuracy.

5. Conclusions

In this paper, a feature reuse (FR) based multi-model fusion method is proposed for SOH estimation of Li-ion batteries. A new health feature, PIVT, is proposed to reflect changes in battery energy. In order to describe the battery aging process more comprehensively, the battery aging process is analyzed from the perspectives of energy, sample entropy and time(micro/macro), and then health features are extracted. To improve robustness, the proposed model is trained using a two-layer structure. The first layer feeds the extracted features into the model separately. The second layer uses the single model-based estimation results and extracted features as input to improve prediction accuracy. KNNR, RFR, ERTR, and LR are fused using BLR. The proposed method is validated by experiments on both Oxford and CALCE datasets. Experimental results show that the RMSE of this method is less than 0.5% and 1% using Oxford and CALCE datasets, respectively.

CRedit authorship contribution statement

Junqi Bai: Conceptualization, Methodology, Software, Validation, Writing – original draft, Writing – review & editing Investigation.
Jiayin Huang: Writing – review & editing, Formal analysis, Project

administration, Visualization. **Kai Luo:** Writing – review & editing, Formal analysis. **Fan Yang:** Writing – review & editing, Data curation. **Yanhua Xian:** Funding acquisition, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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